

Olduvai Gorge in Tanzania is a site of great archaeological significance and is sometimes referred to as the “Cradle of Mankind.” At least two species of early hominins are associated with this area.

THE DIFFERENCES BETWEEN HUMANS AND OTHER APES DNA and blood proteins suggest that our lineage separated from that of the chimpanzees between 8 and 6 million years ago (MYA). Only a few fossil specimens date to this time: *Sahelanthropus tchadensis* (7–6 MYA), *Orrorin*

7 MYA THE TIME WHEN THE FIRST HUMAN ANCESTOR APPEARS

tugenensis (6.1–5 MYA), and two species of *Ardipithecus*, *kadabba* (5.8–5.2 MYA) and *ramidus* (4.4 MYA). While all of these species seem to have **walked on two legs** like us, it is not certain whether any were actual ancestors of humans. Because species are **constantly evolving**, and individuals of those species can vary, it is difficult to tell from isolated and often poorly preserved fossils which species they should be assigned to, or how these are related to one another. However, these fossils do tell us a great deal about what the **last common ancestor** we shared with chimpanzees was like.



Lucy

This unusually complete skeleton of *Australopithecus afarensis*, discovered in Kenya in 1974, was named after the Beatles' song “Lucy in the Sky with Diamonds.”

SEVERAL DIFFERENT AUSTRALOPITHECINE species lived in Africa between 4.2 and 2 MYA. Although they walked on two legs most of the time, they were rather **small and apelike**

and still lived partially in trees. Their brains were about the size of those of modern chimpanzees, but some australopithecines seem to have **used tools**. The earliest stone tools come from Ethiopia and date to **2.6 MYA**, but bones with cut marks made by stone tools have been found associated with *Australopithecus afarensis* nearby, and date to 3.4 MYA. The australopithecines' descendants followed two distinct modes of life: members of the genus *Paranthropus* had huge jaws and big teeth for eating tough vegetable foods; meanwhile, *Homo rudolfensis* and *H. habilis* seem to have eaten **more protein**, using tools to get at the protein-rich marrow inside long-bones by scavenging from carnivore kills.

ULTIMATELY, THE PARANTHROPINES' WAY OF LIFE was unsuccessful and they became **extinct** after about **1.2 MYA**, while their cousins *Homo habilis* and *H. rudolfensis* survived. These early *Homo* species were not very different from australopithecines.

It was with *Homo ergaster* (1.8 MYA) that our ancestors started to **look much more familiar**. *H. ergaster* was tall and slender, and may have been the first hominin (a term used to describe humans and their ancestors) without much body hair. Their **brains were larger** than those of their ancestors, and they lost the last of their adaptations to tree-climbing to become fully adapted to walking and running.

OLDOWAN TOOL



ACHEULEAN TOOL



TOOLS

Many animal species use natural objects as tools, but the manufacture of stone tools is unique to hominins. The earliest are simply sharp flakes broken off stone cobbles by striking them with a “hammerstone.” These are known as “Oldowan” tools, after Olduvai Gorge, where they were first found. Later tools, such as Acheulean handaxes, required more skill. Our manufacture of tools might be one explanation for the evolution of the human brain.

NOT LONG AFTER THE APPEARANCE of *Homo ergaster*, hominins expanded their range **beyond Africa** for the first time. A species called *H. georgicus* appeared in Dmanisi, Georgia, by **1.7 MYA**. Another close relative of *Homo ergaster*, *Homo erectus*, lived in China and Indonesia perhaps not long afterward. Some archaeologists believe that earlier groups of hominins may also have left Africa, as some of the skulls from Dmanisi and from the much later site of Liang Bua in Flores, Indonesia, (currently known as *Homo floresiensis*) resemble those of *Homo habilis* and *Homo rudolfensis*. Living farther north would have required a **different way of life**



5.7–5.2 MYA *Ardipithecus kadabba*, possibly only a variant of *A. ramidus*

7 MYA Hominin and chimpanzee lineages diverge

7–6 MYA *Sahelanthropus tchadensis* may predate hominin/chimpanzee split, but is thought to walk upright

4.4 MYA *Ardipithecus ramidus* (“Ardi”) is chimpanzee-like, but definitely walks upright

4.1–2 MYA *Australopithecus afarensis* has a brain not much larger than that of modern chimpanzees; still habitually climbs trees

3.6 MYA Hominin footprints preserved in volcanic ash at Laetoli in Tanzania

3.5–2 MYA *Australopithecus africanus* the first early hominin to be identified

3.3 MYA Earliest cut marks produced by stone tools on bones at Dikika, Ethiopia

2.6 MYA Earliest known stone tools from Gona, Ethiopia

3.18 MYA “Lucy,” *Australopithecus afarensis*, alive; 13 males and females of varying ages form “first family” group of *A. afarensis* fossils

2.5–1.2 MYA “Nutcracker man,” *Paranthropus boisei*, has huge grinding teeth and powerful jaws

2–1 MYA *Paranthropus* robustus, first paranthropine to be discovered

1.9–1.6 MYA *Homo habilis* (“handy man”) is associated with stone tools and cut-marked bones

1.8 MYA *Homo ergaster* is much taller and more slender than its ancestors

1.7 MYA Earliest known hominin fossils from Eurasia (*Homo georgicus*), from Dmanisi in Georgia

1.65 MYA Acheulean handaxes appear — they may mark a significant step forward in human intelligence